

Alexander B. Chen

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AWARDS & HONORS

Mathematics

United States of America Mathematics Olympiad (USAMO)[†] National · 2026

Scored a 5, top 50% of 250-300 participants was 6

American Invitational Mathematics Examination (AIME)* National · 2024-present

3-time qualifier; top score 12

American Mathematics Competition (AMC) 10/12 National · 2023-present

1st Place in Shady Side Academy for [AMC 12](#) in 2025-2026

1st Place in Shady Side Academy for [AMC 10](#) in 2023-2024 and 2024-25

	2025-26 (Grade 11)	2024-25 (Grade 10)	2023-24 (Grade 9)
AIME I*	12 (top 5% is 13)	8	5
AMC 10/12	117 (12A) 127.5 (12B, top 5%)	112.5 (10A, top 5%) 94.5 (12B)	115.5 (10A, top 5%) 111 (10B)

[†] First-ever [USAMO](#) qualifier from Shady Side Academy

* Highest-ever recorded score on the [AIME](#) from Shady Side Academy

MathLeague High School (Inter)National Championship

1st Place in [MathLeague High School National Championship](#), 11th Grade National · 2026

2nd Place in [MathLeague High School International Championship](#), 10th Grade International · 2025

MathLeague High School State Championship

1st Place in [MathLeague High School State Championship](#), 11th Grade State · 2026

1st Place in [MathLeague High School State Championship](#), 10th Grade State · 2025

United States of America Mathematics Talent Search (USAMTS)

Gold Prize National · 2026

The [USAMTS](#) is a free, national proof-based math competition (consisting of three rounds) open to all U.S. middle and high school students. It gives participants a month to write up detailed solutions to challenging problems and puzzles. Roughly top 5% of all participants receive Gold prizes, the next 7.5% Silver, the following 12.5% Bronze, and finally the next 20% receive Honorable Mentions.

High School Mathematical Contest in Modeling (HiMCM)

Meritorious (M) prize International · 2025

Honorable Mention (HM) prize International · 2024

The [HiMCM](#), run by COMAP, is an international team contest that trades numerical problem-solving for problem-designing. Teams of up to four are tasked with addressing a physical problem by designing a mathematical model and typing up a research-style paper.

- **2024 (Problem A).** A decision-tree expert system classifying sports into families and scoring each on popularity (Olympic YouTube view counts), accessibility, safety, and marketability was built that recommended which sports the International Olympic Committee should add or cut for the 2032 Brisbane Olympics. We used Monte Carlo simulation in Python to generate weighted inputs and logistic-growth curves to project the relevance of fast-emerging sports.
- **2025 (Problem B).** A Pressure-State-Response model ranking Super Bowl host cities by environmental footprint was built that identified Tampa as the most sustainable host. I led the data analysis by using ARIMA for emissions forecasting, AHP-CRITIC for weighting, TOPSIS for city-ranking, and MICE and random-forest imputation for missing indicators, and we were able to generalize our results to the NBA All-Star Weekend.

University of Pittsburgh High School Integration Bee (HSIB)

Champion Regional · 2026

Runner-Up Regional · 2025

Business

Distributive Education Clubs of America (DECA)

DECA prepares high schoolers for careers in business, marketing, finance, and entrepreneurship, and runs a competition ladder from regionals up to the International Career Development Conference. My event was the Innovation Plan (EIP): a team starts from a blank page, invents a genuinely new product or service, works it into a full business plan, and defends it to a panel of judges.

International Career Development Conference (ICDC)

2-time qualifier

International · 2024/25

State Career Development Conference (SCDC)

State champion

State · 2025

3rd Place

State · 2024

- **2024 — CookGPT.** By curating a generative-AI cooking assistant, we attempted to tailor the experience to each user’s specific taste preferences. A single photograph of your fridge allows the app to identify the ingredients and the AI builds recipes from what’s on hand, tracks each meal’s nutrition to flag surpluses and shortfalls, and learns your tastes over time. The pitch was targeted to busy parents, seniors and the elderly, and people with dietary restrictions on a subscription model.
- **2025 — VITALIS.** A wearable health monitor was developed, our main target being aimed at strokes and heart attacks, where continuous biometric sensors feed a cloud database, machine-learning models flag danger patterns and raise preventive alerts, and an integrated microneedle system delivers emergency medication while contacting help. We also offer a plan to sell users’ data to hospitals and research institutions to both better understand patients’ medical conditions as well as making such data more widely accessible. The pitch was targed to high-risk patients, clinicians, and research institutions on a device-plus-subscription model.

Foreign Language

National Latin Exam (NLE)

Latin II, *Maxima Cum Laude*, Silver Medal

National · 2024

Latin III, *Cum Laude*, Honorable Mention

National · 2025

Pittsburgh Chinese School (PCS)

Outstanding Graduate

Regional · 2024

Chinese Cultural Scholarship

Regional · 2023-24

A weekend school for Mandarin language and Chinese culture, taught over years of coursework through to graduation. In 2024, I completed the PCS curriculum and all the requirements, officially graduating with honors. Its Outstanding Graduate honor goes to only two students in each graduating class; I was invited to give a speech during the graduating ceremony.

Music

Pittsburgh International Piano Competition (PIPC)

2nd Place; invited to perform at the Winner’s Concert

International · 2024

Steinway Society of Western Pennsylvania (SSWPA)

3-time winner, and invited to perform at Heinz Hall

Regional · 2024/25/26

Westmoreland Symphony Orchestra (WSO): Young Artist Solo & Ensemble Festival

Gold Prize, scoring 98%

Regional · 2026

Swimming

Eastern Zone Long Course Meters Age Group Championships (EZ LCM AGC)

Competed for the Allegheny Mountain Swimming (AMS) All-Star Swim Team with all states in the Eastern Zone.

Zonal · 2024

Relays

13th Place in the 4×100m Freestyle Relay

13th Place in the 4×50m Freestyle Relay

15th Place in the 4×100m Medley Relay

Individual

100m Freestyle (58.06)

50m Freestyle (27.22)

200m Individual Medley (IM) (2:28.35)

Pennsylvania Interscholastic Athletic Association (PIAA) State Championships

2-time qualifier for the annual PIAA State Championship swim meet at Bucknell University.

State · 2025/26

Placed 6th in the 4×100 Freestyle Relay (2025). Qualified for B-final in the 4×50 Freestyle Relay (2026). (short course yards, SCY)

Western Pennsylvania Interscholastic Athletic League (WPIAL)
3-time qualifier for annual [WPIAL Championship](#) swim meet at the University of Pittsburgh (short course yards, SCY).

	2025-26 (Grade 11)	2024-25 (Grade 10)	2023-24 (Grade 9)
4×50 Freestyle Relay	3rd	—	—
4×50 Medley Relay	4th	—	—
100 Backstroke	10th (57.80)	12th (58.55)	13th (59.73)
200 Freestyle	10th (1:51.07)	—	—
200 Individual Medley	—	15th (2:07.30)	—

Western Pennsylvania Interscholastic Athletic League (PIAA District VII) Section Champions
Shady Side Academy’s first-ever back-to-back undefeated seasons for Boys Varsity Swimming & Diving, earning back-to-back Section Titles/Section Champions. Regional · 2024-25/2025-26

ACTIVITIES

Research & Projects

Program for Research in Mathematics (PReMa) — Texas A&M University

A year of mentored research on Anderson localization and discrete random Schrödinger operators with Prof. Wencai Liu and Dr. Xueyin Wang. First took an intensive college-level semester-long proof-based Linear Algebra course. The project proves exponential decay of the eigenfunction correlator for the one-dimensional Anderson model with a Hoeffding-type error bound, and uses Monte Carlo methods to estimate the localization length across Bernoulli, discrete-uniform, and continuous-uniform disorder. The resulting paper, *Measuring Localization in the One-Dimensional Anderson Model*, is posted as an arXiv preprint and is published at *Advances in Nonlinear Analysis*. Conferences attended:

- Accepted and invited to present at MAA MathFest in Boston
- A talk at the PReMa research seminar

2026

2026

Summer Programs

College in High School – Accelerated High School Program (AHS)

Swimming

Music

LEADERSHIP & VOLUNTEERING

Founder of Shady Side Academy Math Team & Math Club President

Founder & Lead Organizer — Shady Side Academy Mathematics Olympiad (SSAMO)

Pittsburgh Chinese School (PCS)

DaVinci Math AP Calculus Tutor & Teaching Assistant

Greater Pittsburgh Community Food Bank